

CSE321 Theory Assignment 01

Research and Compare Multi-Threading with Event-Driven Concurrency Models Used in Modern Systems such as Node.js by analyzing the following key areas:

- Thread-Based Concurrency
- Event Loop Architecture
- Asynchronous I/O Systems

Report Requirements:

Write a comprehensive analysis focusing on the following key factors:

Thread-Based Concurrency:

- Describe how traditional multi-threaded systems handle concurrent tasks using multiple threads within a process.
- Explain how threads are scheduled by the operating system and how context switching enables concurrent execution
- Discuss the role of synchronization mechanisms such as mutexes, semaphores, and locks in managing shared resources
- Analyze the advantages and limitations of multi-threading, including performance overhead, scalability challenges, and issues like race conditions and deadlocks.

Event Loop Architecture:

- Describe how event-driven architectures operate using a single-threaded event loop.
- Explain how event loops manage task execution using callbacks, event queues, and non-blocking operations.
- Discuss the use of asynchronous programming constructs such as callbacks, promises, and async/await in handling concurrency.
- Analyze how this model avoids thread management complexity while introducing challenges such as blocking operations and debugging difficulty.

Asynchronous I/O Systems and Performance Analysis:

- Define asynchronous I/O and explain how it differs from synchronous I/O in both multi-threaded and event-driven systems.
- Compare how multi-threaded systems and event-driven systems handle I/O-bound and CPU-bound tasks.
- Analyze performance differences under high concurrency workloads, considering factors such as memory usage, scalability, and responsiveness.
- Discuss real-world use cases where each model is more suitable, such as web servers, real-time applications, and computational systems.

Submission guidelines: Can be found in the submission form given below.

Submission link

<https://forms.gle/korX3ccnhRKw2kjT6>

Deadline: 26/4/2026, 11:00 PM

(Submissions outside the Google Form will be automatically rejected.)